HELIX Hybrid Topological PQC Engine

Bare-Metal Benchmark Package (v1.1)

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[1] Helix AI Innovations Inc., Founder — Architecture & Topological Invariants, Azure Standard_E16ads_v7 (Helix-CORE) Base Node

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Target deposition: Zenodo – HELIX Hybrid Topological PQC Engine – Bare-Metal Benchmark Package (v1.1)

Abstract. This release presents bare-metal validation of the HELIX Hybrid Topological PQC Engine and its framing as Proto-AGI: intelligence as a deterministic mesh, not a single bounded model. Primary results were obtained on Helix-CORE (Azure Standard_E16ads_v7, 16 vCPU / 128 GiB). All claims are grounded in kernel traces, native liboqs bindings, and immutable CSV artifacts.

Topological Core. Fixed 8×8 VERIFIED_MATRIX, braid word $[1, 3, 5, 7, -6, 4, -2, 5, 3, 1]$, 8 strands, 10 crossings, Writhe 6, $\mathrm{Tr}(U(B)) = -1+0j$, Mag 1.0, Unitarity Error $7.374\times10^{-9} < 10^{-6}$. Derivation $\sigma = \text{SHA-512}(\mathrm{Tr} \,\|\, B \,\|\, \text{nonce})$, receipt $r = \text{SHA-256}(B \,\|\, \sigma)$.

Entropy & Collision. 1000 nonce-varied seeds, 64 bytes each, 1000 unique, 0 collisions, 7.9970317 bits/byte (99.96% density), 256/256 byte values, chi-square 263.504.

PQC Latency (Helix-CORE). ML-KEM-768 baseline 0.03918 ms, topological 0.02274 ms, overhead −0.01643 ms (−41.95%); ML-DSA-65 baseline 0.09776 ms, topological 0.09077 ms, overhead −0.00699 ms (−7.15%). Native liboqs Python binding.

Kauffman–Jones. State-space 2^c verified: 4 → 16 states 0.013 ms to 14 → 16384 states 4.36 ms.

BKZ Harness — Bilal Khan. Resolved `fpylll_unavailable` to fpylll 0.6.4 live. Deterministic 64-dim lattice from fixed topological matrix, log(determinant) 589.43145, $\log_{10}$ 255.986, $\det^{1/n}$ 9995.262. LLL 2.655 ms, BKZ-20 4.078 ms, BKZ-40 4.515 ms, BKZ-60 4.867 ms, first_vector_norm 9775.774 stable (LLL-optimal, `max_loops=1`, reported as stability).

RHF Correction. Because the basis is diagonal-dominant rather than q-ary, the standard root-Hermite-factor formula yields a raw value < 1. Bilal Khan introduced the publication-safe reporting form $\delta_{\text{corr}} = \max(\delta, 1/\delta)$, producing a corrected RHF of 1.0003525, and replaced the earlier mock value (1.077769) with live bare-metal measurements. Hermite factor 0.97804, determinant via `numpy_slogdet_fallback`. Interpretation: generic integer lattice benchmark, not a formal ML-KEM security proof.

Artifacts. `topology_results.csv`, `entropy_results.csv`, `seed_collision_results.csv`, `mlkem_latency_re` `mldsa_latency_results.csv`, `jones_complexity_results.csv`, `bkz_results.csv`, `rhf_results.csv`, `experiment_metadata.json`

Proto-AGI. Intelligence as mesh consensus across topological invariants, lattice engines, PQC modules, hardware locks, and human authority ($G = 1$). Substrate independence — $\mathrm{Lk} \in \mathbb{Z}$ holds across nodes. The Shape Holds.

Hardware. Helix-CORE — Azure Standard_E16ads_v7 (16 vCPU, 128 GiB, 150k credit track) + GPU burst VMs — Colab Python 3.12.13 reference runs